Three modelling issues found while mapping a TF-PWA model to a scattering-theory standard

$B^+ \to D^- D^{*+} K^+$ — notes for the TF-PWA authors

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As part of the DEMOS effort to map widely-used amplitude-analysis frameworks onto a common standard, we reproduced a TF-PWA model of $B^+ \to D^- D^{*+} K^+$ to numerical precision ($\sim 10^{-10}$ on the complex amplitude, per event) using an independent model builder (CascadeDecays.jl). Achieving that agreement required three constructs that we believe are unintended in TF-PWA. Two are genuine convention/implementation problems that affect the physics; the third is benign for this analysis but is a latent issue elsewhere. This note documents each precisely — with the relevant TF-PWA source locations, the conventional formula, and an indicative (frozen-parameter) impact estimate — so that they can be confirmed, fixed, and re-assessed collaboratively.

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1 Context and scope

This note is part of DEMOS (“Democratizing Models”), an effort to let analysts **share and exchange amplitude models across frameworks**. The vehicle is the `amplitude-serialization` standard: models are expressed in a framework-agnostic form and executed by a reference engine.

The workflow behind the present note is:

1. **A general model builder, grounded in scattering theory.** We use the formalism of arXiv / InspireHEP 2827198, which fixes the helicity/LS conventions, barrier factors, and recoupling unambiguously.
2. **An execution engine.** `CascadeDecays.jl` evaluates such models.
3. **A mapping target.** We map an existing, widely-used **TF-PWA** model of $B^+ \to D^- D^{*+} K^+$ onto (1)+(2).

The TF-PWA model was reproduced **to numerical precision**: the per-event complex amplitude agrees at the $\sim 10^{-10}$ level across the cross-check sample. To reach that agreement, three TF-PWA constructs had to be matched explicitly. We believe all three are unintended, and we summarise them here so they can be discussed and resolved at the source.

We emphasise up front what this note **is** and **is not**. It **is** a precise, reproducible description of three discrepancies between the TF-PWA implementation and the conventional formulas, with code pointers. It is **not** a claim about the final physics impact on the published result: the numbers in Section 6 are *frozen-parameter sensitivities*, and a faithful assessment needs a re-fit (Section 7).

1.1 Decay and topologies

The final state is $D^- D^{*+} K^+$ with $D^{*+} \to D^0 \pi^+$, i.e. four stable tracks D^0, π^+, D^-, K^+ (labelled $1, 2, 3, 4$). Two production topologies appear:

- **D*D resonances** (“DxD”): $B^+ \to X\, K^+$, $X \to D^* D$, with $D^* \to D^0 \pi$. Chain $((((1,2),3),4))$.
- **DK resonances** (“dk”): $B^+ \to D^*\, X$, $X \to DK$, with $D^* \to D^0 \pi$. Chain $(((1,2),(3,4)))$. Here the **second daughter of the B vertex is the spinning resonance X**, and its helicity feeds the $X \to DK$ decay.

This second topology is where Section 3 (the particle-2 phase) becomes observable.

2 Summary of the three issues

Issue A — missing Jacob–Wick particle-2 phase. A static factor $(-1)^{j_2-\lambda_2}$ is omitted at a production vertex whose second daughter is a spinning, further-decaying resonance (frame-alignment / helicity recoupling). This is a convention error that changes a *physical* relative phase between intermediate helicities. It affects the $X_1(2900)$ (DK, $J^P = 1^-$) chain and is the largest of the three effects.

Issue B — non-conventional mass dependence of the multichannel Breit–Wigner. The running width in `BWR_LS` (`ParticleBWRLS`, `split_ls.py`) carries a factor m/m_0 where the conventional phase space gives m_0/m. This is an implementation error that contradicts the model's own docstring; TF-PWA already ships the fix behind a `fix_bug1` flag that defaults to off. It affects the D^*D multichannel states and is minor.

Issue C — running daughter masses in the lineshape denominator. Event-by-event breakup masses are used in the BW running-width denominator where fixed channel masses are appropriate. This does *not* affect the cascade (3-body) setup used here, but is a latent issue for 4-body / floating-mass setups.

The remainder of the note treats each in turn.

3 Issue A — Missing Jacob–Wick particle-2 phase

3.1 Statement

In the Jacob–Wick helicity formalism, a two-body decay $A \to BC$ has the amplitude

$$A^{J_A}_{\lambda_A;\lambda_B\lambda_C} = D^{J_A*}_{\lambda_A,\lambda_B-\lambda_C}(\phi,\theta,0)\,H_{\lambda_B\lambda_C},$$

where the helicity coupling $H_{\lambda_B\lambda_C}$ is built from the LS couplings via Clebsch–Gordan coefficients. Constructing the **second** particle's helicity state requires a rotation that includes a rotation by π, which introduces a static factor

$$\boxed{(-1)^{j_2-\lambda_2}}$$

for daughter 2 (with j_2, λ_2 its spin and helicity; we use the integer-spin convention here, half-integer generalises in the usual way). This factor is part of the standard Jacob–Wick convention and is carried by the builder and by CascadeDecays.

For a **spinless** second daughter, or one whose helicity does not propagate into a subsequent decay, $(-1)^{j_2-\lambda_2}$ is either trivial or an unobservable overall constant. It becomes **physical** when the second daughter is a **spinning resonance that decays further**, because the factor is **helicity-dependent**: it flips sign between $\lambda_2 = +1$ and $\lambda_2 = -1$ for $j_2 = 1$, and λ_2 is summed coherently with the rest of the chain.

In this model the relevant vertex is

$$B^+ \to D^{*+}\,X_1(2900), \qquad X_1(2900) \to D^- K^+,$$

with $X_1(2900)$ a $J^P = 1^-$ state (topology "dk", Section 1.1). Matching TF-PWA required **omitting** $(-1)^{j_2-\lambda_2}$ at this vertex — i.e. TF-PWA does not apply the particle-2 phase, whereas the conventional Jacob–Wick recoupling does.

3.2 How we verified it

The phase is not ad hoc: it is the imprint of the **particle-2 rest-frame transformation**. For the topology $((1,2),(3,4))$, the rotation that takes the B frame into the $(3,4)$ (here $X_1(2900)$) rest frame is, with (θ_1,ϕ_1) the $(1,2)$ (D^*) direction,

$$p \triangleright R_z(-\phi_1) \triangleright R_y(-\theta_1) \triangleright R_y(\pi)$$

read left to right (first $R_z(-\phi_1)$, then $R_y(-\theta_1)$, then the closing $R_y(\pi)$). That closing $R_y(\pi)$ — present because $(3,4)$ recoils *opposite* to $(1,2)$ — acts on a spin-j_2 helicity state as $d^{j_2}_{\lambda,\lambda_2}(\pi) = (-1)^{j_2-\lambda_2}\,\delta_{\lambda,-\lambda_2}$, supplying exactly the $(-1)^{j_2-\lambda_2}$ factor (together with the $\lambda_2 \to -\lambda_2$ flip) for the second daughter.

CascadeDecays includes this $R_y(\pi)$ and so applies $(-1)^{j_2-\lambda_2}$ exactly once. To reproduce TF-PWA we had to cancel it for the three $X_1(2900)$ chains; with that cancellation the full-model amplitude matches TF-PWA at the 10^{-10} level. The $X_0(2900)$ chain ($J^P = 0^+$, $\lambda_2 \equiv 0$) needs no such treatment, consistent with the phase being trivial for a spinless second daughter.

On the TF-PWA side, sequential helicity frames are reconciled in `HelicityDecay.add_algin` (`tf_pwa/amp/core.py`), which rotates each spinning inner particle by its aligned angle through a pure Wigner-D matrix:

```
# tf_pwa/amp/core.py  -  HelicityDecay.add_algin (abridged)
if self.aligned:
    for j, particle in enumerate(self.outs):
        if particle.J != 0:
            ang = data[particle].get("aligned_angle", None)
            ...
            dt = get_D_matrix_lambda(
                ang, particle.J,
                self.list_helicity_inner()[j], particle.spins,
            )
            ...
            ret = dt * ret                          # rotation only - no (-1)^(J - lambda)
            ret = tf.reduce_sum(ret, axis=j + 2)
return ret
```

We see no explicit $(-1)^{j_2-\lambda_2}$ factor for the recoil / second daughter here or in the chain contraction, consistent with it being absent. We would value the authors confirming the precise location and intent.

3.3 Why it is not "just an overall sign"

A common reaction is that this can be absorbed into the resonance's complex coupling. It cannot, in general:

- An overall sign/phase is **one** factor multiplying the whole chain, independent of helicity.

- $(-1)^{j_2-\lambda_2}$ is **helicity-dependent**: it weights the $\lambda_2 = +1$ and $\lambda_2 = -1$ contributions with *opposite* signs before they are summed coherently with the D^* helicities. This changes the predicted angular distribution and the interference pattern of the $X_1(2900)$ chain with every other component. It cannot be reproduced by any single rephasing of the coupling.

4 Issue B — Non-conventional mass dependence in the multichannel Breit–Wigner

4.1 Statement

The D^*D resonances with two LS channels (e.g. S- and D-wave) use TF-PWA's multichannel lineshape `BWR_LS` (`ParticleBWRLS`, `tf_pwa/amp/split_ls.py`). Its **docstring** specifies the conventional running width with two-body phase space $\rho = 2q/m$:

$$R_i(m) = \frac{g_i}{m_0^2 - m^2 - i\, m_0\, \Gamma_0\, \dfrac{\rho}{\rho_0} \sum_i g_i^2}, \qquad \frac{\rho}{\rho_0} = \frac{2q/m}{2q_0/m_0} = \frac{q}{q_0}\,\frac{m_0}{m}.$$

The conventional factor is therefore $\dfrac{m_0}{m}$. The implementation, however, applies $\dfrac{m}{m_0}$ unless an opt-in flag is set:

```
# tf_pwa/amp/split_ls.py  -  ParticleBWRLS.get_ls_amp_frac
a = m0 * m0 - m * m
b = m0 * g0 * tf.sqrt(q2 / q02) * sum([i * i for i in total_gamma])
if self.fix_bug1:
    b = b * m0 / m      # conventional  (matches the docstring: rho = 2q/m)
else:
    b = b * m / m0      # DEFAULT       (inverted mass dependence)
dom = tf.complex(a, -b)
```

So with the default `fix_bug1 = False` the running-width term carries $\dfrac{m}{m_0}$ rather than the documented $\dfrac{m_0}{m}$. The two differ by a factor $\left(m/m_0\right)^2$, i.e. an **extra power of** $s = m^2$ in the width term. The name of the guard flag (`fix_bug1`) indicates this is already recognised internally as a bug; it simply defaults to the unfixed branch, and the model under study was built with the default.

The two branches coincide **at the pole** $m = m_0$ (so Γ_0 at resonance is unchanged) and differ only in the **off-resonance tail**. The "fixed" lineshape used for the impact study below is exactly `fix_bug1 = True`.

5 Issue C — Running daughter masses in the lineshape denominator

5.1 Statement

When an intermediate two-body system has a resonant lineshape with a running width, the breakup momentum $q(m)$ in the **denominator** (running width) is a function of the *running* invariant mass

m of that system, but the **daughter** masses entering q should be the appropriate fixed masses for the channel, not the event-by-event invariant masses of the observed final-state particles.

The reason is a factorisation of the mass dependence of a resonant contribution into two physically distinct pieces:

$$A(s) \;=\; P(s) \,\cdot\, V(s, m_1^2, m_2^2), \qquad s = m^2.$$

- $P(s)$ is the **propagator** — a property of the resonance *itself*. It describes how the particle propagates and "bubbles" through its intermediate states, i.e. the dressing by the self-energy Σ: its denominator is $m_0^2 - s - \Sigma(s)$, of which the running-width term $i\, m_0 \Gamma(s)$ is the imaginary part. Σ is summed over the resonance's channels, whose **thresholds are fixed constants**; P is therefore a function of s alone.
- $V(s, m_1^2, m_2^2)$ is the **vertex / form factor** for the *particular* decay into the observed final state of masses m_1, m_2. This is where the final-state kinematics legitimately live — the breakup momentum and barrier factor of that specific channel.

The BW denominator is part of P, **not** V. It must therefore depend on s only, with the channel masses entering Σ at their fixed values; it should **not** carry the event-by-event daughter masses of the observed final state. Using running daughter masses in the denominator mixes the vertex kinematics of V into the propagator P. In a **cascade** phase-space setup the relevant daughter (here D^*) is held at its nominal mass, so P's argument is unaffected and this never bites. In setups where the daughter mass itself **floats** — e.g. flat 4-body phase space, where $m(D^0\pi)$ varies event by event and is not pinned to $m(D^*)$ — TF-PWA's use of event-by-event breakup masses in the running-width denominator produces a different lineshape than the conventional fixed-mass prescription.

5.2 Why it does not affect this analysis

The production B2DxDK setup uses **cascade** phase space with the D^* fixed at its nominal mass. With $m(D^0\pi) \equiv m_{D^*}$, the nominal and running daughter masses coincide and the issue is invisible. We surfaced it only via a flat 4-body cross-check: after switching the BW propagators to event breakup masses consistently, the independent builder and TF-PWA agree to $\sim 10^{-10}$ on a 1000-event flat-4b sample. We include it here for completeness and because it is a **latent** issue for any analysis in which intermediate masses float under the sampled phase space while multichannel/running-width lineshapes are used.

6 Indicative impact on B2DxDK

Method. We rebuild the production cascade with one issue "fixed" at a time, **freezing all fitted parameters**, and recompute the weighted fit fractions on the full 10^5-event sample (`data/b-decay-events.arrow`). Issue C is excluded by construction (cascade phase space). The baseline reproduces the committed TF-PWA fit fractions. Reported quantities are the baseline fit fraction and the *changes* under each fix (percentage points). The $\pm$ on the baseline is the MC statistical uncertainty: each per-event weighted contribution $y_{k,i} = w_i |A_{k,i}|^2$ is a sample mean, so the row sum $N_k = \sum_i y_{k,i}$ has uncertainty $\sigma(N_k) = \sqrt{n}\,\mathrm{std}(y_{k,\cdot})$; dividing by the coherent total D (whose own uncertainty we neglect) gives $\sigma(\mathrm{FF}_k) = \sigma(N_k)/D$. The ΔFF columns are evaluated on the *same* events for both models, so they are statistically far more precise than the absolute fractions (the MC fluctuations largely cancel in the difference) and no separate error is quoted on them.

- ΔFF_A: fix A only (restore the Jacob–Wick particle-2 phase on $X_1(2900)$).
- ΔFF_B: fix B only (`fix_bug1 = True` mass dependence on the multichannel D^*D states).
- ΔFF_{AB}: both.

Table 1: Baseline fit fractions and their changes under each fix. The quoted $\pm$ on the baseline are **statistical uncertainties only**, propagated from the Monte-Carlo computation on the 10^5 (100,000)-event phase-space sample; the ΔFF columns, evaluated on the same events, are statistically far more precise (see text).

Component	base (%) $\pm$ stat	ΔFF_A (pp)	ΔFF_B (pp)	ΔFF_{AB} (pp)
X(3872)	10.94 ± 0.17	-0.81	-0.13	-0.92
X(3915)(0−)	3.35 ± 0.03	-0.25	-0.04	-0.28
chi(c2)(3930)	1.77 ± 0.04	-0.13	-0.02	-0.15
X(3940)(1.)	5.04 ± 0.04	-0.38	$+0.15$	-0.23
X(3993)	9.90 ± 0.15	-0.74	-0.08	-0.79
Psi(4040)	2.80 ± 0.03	-0.21	-0.03	-0.23
X(4300)	1.16 ± 0.01	-0.09	-0.01	-0.10
NR(0−)SPp	15.81 ± 0.07	-1.18	-0.19	-1.33
NR(1.)PSp	17.59 ± 0.05	-1.31	-0.21	-1.47
NR(0−)SPm	1.16 ± 0.004	-0.09	-0.01	-0.10
NR(1−)PPm	20.41 ± 0.07	-1.52	-0.25	-1.71
X0(2900)	6.51 ± 0.05	-0.48	-0.08	-0.55
X1(2900)	5.50 ± 0.04	-0.42	-0.07	-0.47
interference	-1.94 ± 0.31	$+7.59$	$+0.97$	$+8.32$

Reading the numbers.

- **Issue A is the dominant effect.** Restoring the particle-2 phase on the single $X_1(2900)$ chain swings the **total interference by** $+7.6$ **pp** (from -1.9% to $+5.7\%$): the model becomes markedly more constructive. Because interference is global, *every* component fit fraction moves — up to 1.5 pp — even though only one chain changed. This confirms the expectation that A "might be impactful".
- **Issue B is minor** at frozen parameters: $\lesssim 0.25$ pp per component and ~ 1 pp on interference, consistent with the expectation of a small effect. (It is nonetheless a real shape change in the D^*D lineshapes and should be fixed.)
- **Issue C** does not enter the cascade analysis.

Crucial caveat. These are **frozen-parameter sensitivities**, not the physics impact. The fit fractions are self-normalising, so the uniform downward shift of all components under fix A reflects the increased coherent total; with the parameters held fixed, the corrected model is simply a *different* model. The genuine impact — on fitted parameters, fit fractions, and their uncertainties — can only be obtained by **re-fitting**, where the couplings will absorb part of the change. The numbers above should be read as *"this is how much the model moves if nothing else adapts"*, i.e. an upper-bound-flavoured sensitivity rather than a prediction of the post-fit shift.

7 Proposed collaborative follow-up

We would like to resolve these at the source, with the TF-PWA authors, rather than only working around them downstream. Concretely, for each issue we suggest the loop **(a) confirm → (b) fix → (c) re-assess on the original data with the original setup**:

1. **Confirm.**
   - *A:* agree on whether TF-PWA intends to omit the Jacob–Wick particle-2 phase for a spinning, further-decaying second daughter, and pin the exact code location (frame alignment in `tf_pwa/amp/core.py`).
   - *B:* confirm that `ParticleBWRLS` default (`fix_bug1=False`) contradicts its own docstring ($\rho = 2q/m \Rightarrow m_0/m$), and that the published model was built with the default.
   - *C:* confirm the intended daughter-mass prescription in the running-width denominator when intermediate masses float.
2. **Fix.** For B the fix already exists (`fix_bug1=True`); for A a convention-consistent particle-2 phase would be added; for C a fixed-mass prescription in the denominator.
3. **Re-assess on the original data with the original setup.** Re-fit the published model after each fix and compare fitted parameters, fit fractions, and their uncertainties. This is the step that turns our indicative sensitivities into a real impact statement, and it is best done by the authors on the original inputs. We are happy to cross-check every step against the independent builder.

8 Reproduction

All artefacts referenced here live in the B2DxDK repository:

- Independent model (TF-PWA-aligned): `src/` (`lineshapes.jl`, `recoupling.jl`, `model.jl`, `matching.jl`, `resonance_table.jl`).
- Per-event 10^{-10} amplitude cross-check: `test/runtests.jl` against `data/crosscheck_amplitudes_referenc`
- Baseline fit fractions: `scripts/all_resonances_fit_fractions.jl`.
- The impact study in Section 6: `scripts/abc_impact.jl` (rebuilds the cascade with each fix toggled and recomputes fit fractions, with statistical uncertainties).
- TF-PWA source referenced above is the pinned submodule under `archive/investigation/tf-pwa/` (e.g. `tf_pwa/amp/split_ls.py`, `tf_pwa/amp/base.py`, `tf_pwa/amp/core.py`, `tf_pwa/breit_wigner.py`).

The flat-4b cross-check that exposed issue C is archived under `archive/flat4b/`.